$$\text{Therefore, } f'(c) = 0 \text{ gives } c = 0,1, \text{ and } \frac{1}{2}$$

$$\text{which } \Rightarrow c = \frac{1}{2} \in (0,1).$$

Example 7.20

Find the value in the interval $\left(\frac{1}{2},2\right)$ satisfied by the Rolle's theorem for the function $f(x) = x + \frac{1}{x}, x \in \left[\frac{1}{2},2\right]$.

Solution

We have, $f(x)$ is continuous in $\left[\frac{1}{2},2\right]$ and differentiable in $\left(\frac{1}{2},2\right)$ with $f\left(\frac{1}{2}\right) = \frac{5}{2} = f(2)$. By the Rolle's theorem there must exist a value $c \in \left(\frac{1}{2},2\right)$ such that,

$f'(c) = 1 - \frac{1}{c^2} = 0 \Rightarrow c^2 = 1$ gives $c = \pm 1$. As $1 \in \left(\frac{1}{2},2\right)$, we choose $c = 1$.

Example 7.21

Compute the value of 'c' satisfied by Rolle's theorem for the function $f(x) = \log\left(\frac{x^2+6}{5x}\right)$ in the interval $[2,3]$.

Solution

Observe that, $f(2) = 0 = f(3)$ and $f(x)$ is continuous in the interval $[2,3]$ and differentiable in $(2,3)$. Now,

$$f'(x) = \frac{x^2-6}{x(x^2+6)}$$

$$\text{Therefore, } f'(c) = 0 \text{ gives}$$

$$\frac{c^2-6}{c(c^2+6)} = 0$$

$$\text{which implies } c = \pm\sqrt{6}$$

$$\text{Now } c = +\sqrt{6} \in (2,3).$$

Observe that $-\sqrt{6} \notin (2,3)$ and hence $c = +\sqrt{6}$ satisfies the Rolle's theorem.

Rolle's theorem can also be used to compute the number of roots of an algebraic equation in an interval without actually solving the equation.

Example 7.22

Without actually solving show that the equation $x^4 + 2x^3 - 2 = 0$ has only one real root in the interval $(0,1)$.

Solution

Let $f(x) = x^4 + 2x^3 - 2$. Then $f(x)$ is continuous in $[0,1]$ and differentiable in $(0,1)$. Now,

$$f'(x) = 4x^3 + 6x^2 \text{. If } f'(x) = 0 \text{ then,}$$

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